TEXORA · LECTURE SHEET

Consumption and Costing

How fabric and trim consumption is calculated, how a garment cost sheet is built line by line, what decides the final price, and how cost differs from price — with two fully worked cost sheets: the course woven shirt and a knit T-shirt costed by the panel method.

TE-4104 · APPAREL MERCHANDISING AND INTERNATIONAL TRADE · TOPIC 4 · LEVEL 4
TERM 1

1. The system of fabric consumption

Fabric consumption is the quantity of fabric required to make one garment (or one dozen), calculated from the garment's measurements / pattern and the usable fabric width, with allowances added. Costing must use **gross consumption** — the fabric actually bought per garment — never the net area of the finished panels.

From net to gross

1. **Panel / marker calculation.** Lay the garment's panels on the fabric width (by marker, or by formula for knit panels — Section 5) to find the fabric length or weight the panels occupy: the *net* consumption.
2. **Add process allowances.** Shrinkage allowance (fabric will shrink in washing / finishing), cutting wastage (marker ends, defects, selvedge loss), and a small production wastage allowance.
3. **Gross consumption** = net consumption increased by the total allowance percentage. Fabric is booked and costed at gross.

$$\text{Gross consumption} = \text{Net consumption} \times (1 + \text{total wastage \%})$$

Two measurement systems are used. **Woven fabric** is bought by length, so consumption is expressed in metres (or yards) per piece at a stated width. **Knit fabric** is bought by weight, so consumption is expressed in kilograms per piece (or per dozen), derived from panel area × GSM.

2. Garment costing: the structure

A **cost sheet** assembles every cost of executing the order, per piece:

$$\text{Total cost} = \text{Fabric} + \text{Trims \& accessories} + \text{CM} + \text{Process additions} + \text{Testing \& inspection} + \text{Overhead \& commercial}$$

FOB price = Total cost + Profit (margin on total cost)

- **Fabric cost** = gross consumption × fabric price (per metre for woven, per kg for knit).
- **CM (cut and make)** = the conversion charge for cutting, sewing and finishing, derived from the factory's cost per minute × the garment's standard minute value, or quoted as a conversion rate.
- **Process additions** — washing, printing, embroidery and similar — are separate lines wherever they apply, never hidden inside CM.
- **Profit** is applied last, on the whole cost, because the business commits its capital and risk to the whole order.

3. Factors influencing price determination

FACTOR	HOW IT MOVES THE PRICE
Fabric price and consumption	The dominant line: a wider usable width or a better marker lowers consumption; a higher fabric rate raises cost directly.
Order quantity	Fixed costs of development, set-up and documentation are spread over more pieces in a large order, lowering cost per piece.
Style complexity	More operations, matching, special processes and tighter tolerances raise CM and wastage.
Quality and testing level	Higher performance standards, more tests and third-party inspection add cost.
Lead time	A compressed schedule may force air freight of materials, overtime or premium bookings.
Payment and delivery terms	Longer credit, delivered-duty terms and currency risk are priced into the offer (Topic 5).
Market competition	Competing factories and countries set the band within which a negotiated price must fall (Topic 1).

4. Trims and accessories consumption

Trims are costed item by item from the BOM (Topic 2), each with its own consumption basis: sewing thread by estimated length per garment (by garment type and seam length); buttons, rivets and snaps by count per garment plus a wastage allowance for breakage and rejection; zips by count and length; interlining by area (metres at a stated width, like a second fabric); elastic and tape / ribbon by length per garment; labels, hangtags and price tickets by count; polybags by count; cartons by the number of

pieces packed per carton (carton cost ÷ pieces per carton = cost per piece). As with fabric, trim consumption used in costing is **gross** — count plus wastage — because broken buttons and rejected labels are bought whether or not they reach a garment.

5. Direct and indirect cost; costing vs. pricing

CLASS	DEFINITION	GARMENT EXAMPLES
Direct cost	Traceable to the individual garment / order.	Fabric, trims, CM labour for the style, style-specific processes (wash, print), packing for the order.
Indirect cost	Necessary to run the business but not traceable to one garment; recovered through overhead.	Factory rent and utilities share, management and office salaries, banking and documentation charges, transport to port.

Costing vs. pricing. Costing is the internal, factual build-up of what an order costs — it answers to arithmetic. **Pricing** is the external decision of what to charge — it starts from the cost, then considers the buyer's target, competition, the value of the relationship and the risk of the terms. Cost is a floor, not a price: quoting below cost loses money on every piece, and quoting cost-plus without reading the market can price a factory out of an order it could profitably win at a keener margin. The cost sheet disciplines both errors.

6. Worked cost sheet A — woven shirt (course order)

Order: **12,000 pieces** of a woven shirt. Fabric: cotton poplin, GSM 120, usable width 1.50 m, price **$3.20 per metre**. Gross consumption from the marker: **0.927 m per piece**.

Fabric cost = 0.927 m × $3.20/m = $2.97 per piece (rounded)

COST ELEMENT	BASIS	US$ / PIECE
Fabric	0.927 m (gross) × $3.20/m	2.97
Trims & accessories	Thread, buttons, interlining, labels, polybag, carton share	0.40
CM (cut & make)	Cutting, sewing and finishing conversion charge	0.63
Testing & inspection	Fabric / garment tests and final inspection, amortised	0.10
Overhead & commercial	Banking, documentation, transport to port, miscellaneous	0.18

Total cost		**4.27**
Profit 10%	10% on total cost	0.43
FOB price		**4.70 / piece · 56.38 / dozen**

Figures are the established course-shirt figures, presented rounded to cents as quoted to the buyer. Fabric dominates the cost — about seven dollars in every ten — which is why consumption accuracy (Topic 2's tech pack, the marker, the width) matters more than any other single number on the sheet.

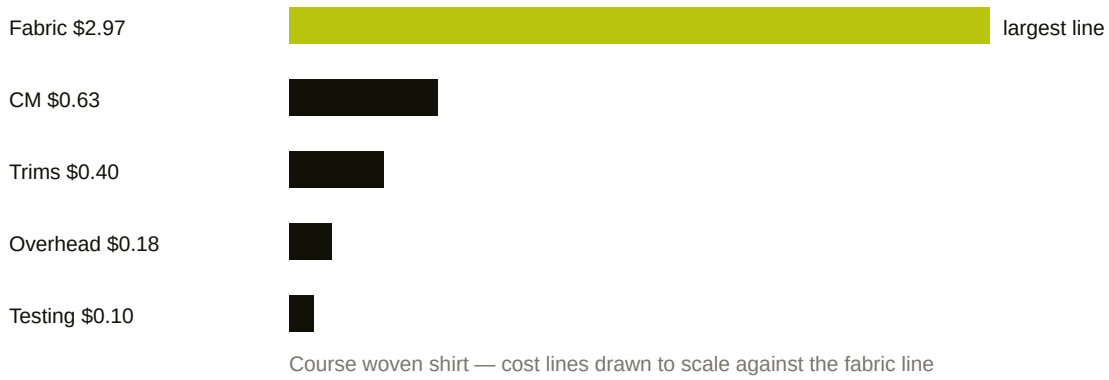


Figure 1 — Cost structure of the course woven shirt (total cost $4.27/pc, FOB $4.70/pc). The fabric bar is the lesson: control consumption and fabric price first.

7. Worked cost sheet B — knit T-shirt by the panel method

For knit garments, consumption is calculated by weight from the panels. In the simplified panel method used here, body and sleeve are treated as one combined panel whose length is (body length + sleeve length) and whose width is the full chest:

$$\text{Net consumption (g/pc)} = (\text{Body length} + \text{Sleeve length}) \times \text{Full chest} \times \text{GSM} \div 10{,}000$$

[lengths in cm]

Inputs (stated in full): body length 70 cm; sleeve length 20 cm; full chest 100 cm; fabric single jersey, GSM 160; wastage allowance 10%; fabric price $6.00 per kg.

1. Combined length = 70 + 20 = 90 cm. Panel area = 90 × 100 = 9,000 cm² = 0.90 m².
2. Net consumption = 9,000 × 160 ÷ 10,000 = **144 g per piece**. (Check by area: 0.90 m² × 160 g/m² = 144 g. ✓)
3. Gross consumption = 144 × 1.10 = **158.4 g = 0.1584 kg per piece**.
4. Fabric cost = 0.1584 kg × $6.00/kg = **$0.9504 ≈ $0.95 per piece**.

COST ELEMENT	BASIS	US$ / PIECE

Fabric	0.1584 kg (gross) × $6.00/kg	0.95
Trims & accessories	Thread, neck rib share, labels, polybag, carton share	0.25
CM (cut & make)	Cutting, sewing and finishing conversion charge	0.45
Testing & inspection	Tests and final inspection, amortised	0.05
Overhead & commercial	Banking, documentation, transport to port	0.10
Total cost	0.95 + 0.25 + 0.45 + 0.05 + 0.10 (unrounded fabric $0.9504)	**1.80**
Profit 10%	10% on total cost	0.18
FOB price		**1.98 / piece · 23.76 / dozen**

Arithmetic check on unrounded figures: total cost = 0.9504 + 0.25 + 0.45 + 0.05 + 0.10 = $1.8004; FOB = 1.8004 × 1.10 = $1.9804, quoted as $1.98 per piece, $23.76 per dozen at the quoted price. The method note matters as much as the numbers: a real marker-based calculation separates body, sleeve and rib panels; the combined-panel method is the quick estimating form, and the sheet must state which method produced its consumption.

8. Cost-sheet outlines for the other listed garments

GARMENT	CONSUMPTION INPUT AND METHOD	COST-SHEET NOTES
Polo shirt	Knit panel method as Section 7, with the collar and placket panels added; piqué fabric GSM and price per kg.	Trims line carries collar / cuff (knitted to size), placket and buttons; CM higher than a T-shirt for placket and collar setting.
Hoody	Knit panel method for body, sleeves and hood panels; fleece fabric by weight; drawcord and eyelets by count / length.	Fabric weight is the dominant line (heavy GSM); trims include drawcord, eyelets, rib for cuff and hem; CM includes hood construction.
Pant (woven)	Marker consumption in metres at stated width, as for the woven shirt, from waist / hip and inseam measurements.	Trims: zip, button, hook, pocketing fabric, waistband materials; a wash line is added where the style is washed; CM reflects fly and waistband operations.
Skirt	Marker consumption in metres at stated width from skirt length and hem sweep;	Trims: zip / hook, waistband, lining; CM lower than pants, rising with pleats, slits and

	lining consumption costed separately where lined.	lining work.
Sweater	Knitted-to-shape panels costed by garment weight: finished garment weight plus knitting / linking wastage, × yarn price per kg.	Yarn is the fabric line; CM covers linking, mending and finishing; trims are minimal (labels, buttons where cardigan style).
Jacket	Marker consumption in metres for shell fabric, plus separate consumption for lining and interlining / padding.	The fullest sheet: three fabric-type lines, the largest trims list (zip, buttons, snaps, drawcords), the highest CM — every addition shown as its own line.

Every outline follows the same skeleton as Sections 6 and 7 — consumption first, then trims by their own bases, CM, additions as separate lines, testing, overhead, profit last. Only the inputs and the risk points change.

Glossary

TERM	MEANING
Consumption	The fabric (length or weight) or trim quantity required per garment; gross consumption includes all allowances.
Cost sheet	The itemised build-up of a garment’s total cost and FOB price.
CM	Cut-and-make charge: the conversion cost of cutting, sewing and finishing.
Direct / indirect cost	Cost traceable to one garment / order, vs. general business cost recovered through overhead.
Pricing	The decision of what to charge, made from the cost plus market, terms and relationship considerations.
GSM	Grams per square metre: the weight basis for knit fabric consumption by the panel method.

Key points and summary

- Cost at gross consumption always: net panel quantity plus shrinkage, cutting and production allowances.
- Woven consumption is in metres at a stated width; knit consumption is in kilograms from panel area × GSM.

- Total cost = fabric + trims + CM + separate process additions + testing + overhead; profit is applied last, on the whole cost.
- Course woven shirt: fabric $2.97, total cost $4.27, FOB $4.70/pc ($56.38/dozen) on a 12,000-piece order.
- Worked T-shirt: net 144 g, gross 158.4 g, fabric $0.95, total cost $1.80, FOB $1.98/pc.
- Costing is fact; pricing is judgement built on that fact. Direct costs trace to the garment; indirect costs are recovered through overhead.

Practice

Answer in your own words; brief model answers follow. Show all working for numerical questions.

1. Using the panel method of Section 7, calculate the net and gross consumption (10% wastage) of a T-shirt with body length 72 cm, sleeve length 22 cm, full chest 104 cm and GSM 180. Then calculate its fabric cost at $6.50/kg.
2. For the course woven shirt, the fabric price rises to $3.60/m, all else unchanged (use the rounded presentation figures). Recalculate total cost and FOB at 10% profit.
3. A carton costing $1.20 holds 24 pieces. What is the carton cost per piece, and in which cost-sheet line does it belong?
4. Classify as direct or indirect: (a) interlining for one shirt order; (b) the factory office's monthly rent; (c) the wash charge for one denim order.
5. Why must a wash be shown as its own cost-sheet line rather than added silently into CM?

Answer key

1. Combined length = 72 + 22 = 94 cm. Net = 94 × 104 × 180 ÷ 10,000 = 175.968 g ≈ 176.0 g. Gross = 175.968 × 1.10 = 193.5648 g = 0.1936 kg. Fabric cost = 0.1935648 × 6.50 = $1.258 ≈ $1.26/pc.
2. Fabric = 0.927 × 3.60 = $3.3372 ≈ $3.34. Total cost = 3.34 + 0.40 + 0.63 + 0.10 + 0.18 = $4.65. FOB = 4.65 × 1.10 = $5.11/pc (using unrounded fabric, $5.11/pc to the cent).
3. $1.20 ÷ 24 = $0.05 per piece; it belongs in trims & accessories (packing materials share).
4. (a) Direct. (b) Indirect — recovered through overhead. (c) Direct — a process addition line for that order.
5. Because washing applies only to washed styles and is priced per order by an outside process; burying it in CM hides its cost, corrupts comparison between washed and unwashed styles, and makes the sheet indefensible line by line.